Optimal Extensions of Cross-Sections: Sphere Packings in Dimensions 38 to 43

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Abstract

We improve the best known sphere packings in every dimension from 38 to 43. Our packings in dimensions 38 to 42 come from one chain of cross-sections of the extremal even unimodular lattice P_{48p}, $\sqrt{3}E_6 \subset \sqrt{3}E_7 \subset \sqrt{3}E_8 \subset K_9 \subset K_{10}$, whose first three members are cut from the fixed lattice $\sqrt{3}(E_8 \perp E_8)$ of an order-three automorphism; their orthogonal complements are lattice packings and set the records in dimensions 42 down to 38. Each successive section is a determinant-minimal extension of its predecessor. Our 43-dimensional packing is an antipode packing: six translates of the complement of a 5-dimensional section.

We also improve some kissing numbers. Conway and Sloane's twelve 1982 cross-section packings appear never to have had theirs computed; we compute them and find that, in dimensions 42 to 47, they exceed the previously tabulated lower bounds. Our chain does better in dimensions 40, 41 and 42, and a further antipode packing beats the record in dimension 45.

1 Introduction

A *sphere packing* in $\mathbb{R}^n$ is a collection of congruent balls with disjoint interiors, and its *density* is the fraction of space the balls cover. Following Conway and Sloane [9], we measure packings by their *center density*

$$\delta(\mathcal{P}) \;=\; \frac{\Delta(\mathcal{P})}{V_n},$$

where $\Delta(\mathcal{P})$ is the density and V_n is the volume of the unit ball. The sphere packing problem is solved only in dimensions 1, 2, 3 [13], 8 [25] and 24 [6]; in every other dimension the state of the art is a record set by an explicit construction, and we refer to the table maintained by Cohn [5].

Around dimension 48 the densest packings known come from the extremal even unimodular lattices of that dimension and their cross-sections. In *Laminated lattices*, Conway and Sloane [7, Corollary 8] exhibited cross-sections of P_{48p} that set the records in dimensions 39–43, and those records stood for four decades; the record in dimension 38 was held by a lattice of another family, D_{38} of center density 8 [9, p. xxxvii]. The antipode construction of Conway and Sloane [8], which places several translates of a cross-section over a cluster in its dual, improved dimensions 44–47; Chen et al. [3] applied it to deliberately suboptimal cross-sections and improved dimensions 19, 20, 21, 23, 44, 45 and 47.

This paper improves all six dimensions from 38 to 43 (Table 1), and, more to the point, derives them from a single mechanism rather than six separate tricks.

The mechanism. Let Λ be one of the four known extremal even unimodular 48-dimensional lattices, so $\det\Lambda = 1$ and the minimal squared norm is $\mu = 6$. A saturated cross-section $K \subset \Lambda$ of rank k has an orthogonal complement $L = \Lambda \cap \operatorname{span}(K)^\perp$ of rank $l = 48 - k$ with $\det L = \det K$ and minimum at least 6,

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	previously known			this paper		
dim	construction	lattice?	δ	construction	lattice?	δ
38	D_{38} [9]	yes	$2^3 = 8$	Section 3	yes	$2^{-19}3^{31/2}5^{-1} = 9.48\ldots$
39	cross-section [7]	yes	$2^{-41/2}3^{16}7^{-1/2} = 10.97\ldots$	Section 3	yes	$2^{-20}3^{16}5^{-1/2} = 18.35\ldots$
40	cross-section [7]	yes	$2^{-45/2}3^{17} = 21.77\ldots$	Section 3	yes	$2^{-20}3^{16} = 41.05\ldots$
41	cross-section [7]	yes	$2^{-43/2}3^{17} = 43.54\ldots$	Section 3	yes	$2^{-21}3^{17} = 61.57\ldots$
42	cross-section [7]	yes	$2^{-22}3^{18} = 92.36\ldots$	Section 3	yes	$2^{-21}3^{35/2} = 106.65\ldots$
43	cross-section [7]	yes	$2^{-45/2}3^{19} = 195.94\ldots$	Section 4	no	$2^{-223/2}3^{-21}137^{43/2} = 226.51\ldots$

Table 1: Center densities δ. The columns "lattice?" say whether the packing is a lattice packing; our dimension-43 packing is an antipode packing of $s = 6$ translates of a lattice in (1).

and a finite set S of s points of the dual lattice K^* of pairwise squared distance at most β produces, by the *antipode construction* of Conway and Sloane [8], an l-dimensional packing of center density

$$\delta \;=\; \frac{s}{2^l}\,\frac{(6-\beta)^{l/2}}{\sqrt{\det K}}. \tag{1}$$

Everything in this paper is a choice of (K, S) in (1), and the two ways to make δ large pull against each other: a small $\det K$ makes K^* sparse, which limits s, while a cluster large enough to pay for a bigger determinant needs a K^* with many short vectors.

Formula (1) and the two bounds that constrain it are due to Conway and Sloane, and are collected in Section 2. This paper adds four things. First, a sharpening of their determinant bound for one-vector extensions. Write $\rho_{\text{glue}}(K)$ for the largest distance from a point of K^* to K, a maximum over the $\det K$ classes of K^*/K; we call it the *glue radius*. Then every rank-$(k+1)$ cross-section containing K has determinant at least $\det K\,(6-\rho^2_{\text{glue}}(K))$, in *every* integral ambient of minimum 6 (Lemma 13). Conway and Sloane control the analogous quantity by a subcovering radius and then by the ordinary covering radius; here integrality restricts the relevant projections to K^*, giving $\rho_{\text{glue}}(K) \le \rho_{\text{cov}}(K)$, the ordinary covering radius, and a finite computation of rank k. Second, the exact values of ρ^2_{glue} along the chain $\sqrt{3}E_6 \;\subset\; \sqrt{3}E_7 \;\subset\; \sqrt{3}E_8 \;\subset\; K_9 \;\subset\; K_{10}$ of cross-sections of P_{48p}, which prove each one the smallest-determinant extension of the one before it (Theorem 14). Third, the sections K_9 and K_{10} themselves, whose complements are the records in dimensions 39 and 38; the first three sections of the chain come from the fixed lattice $\sqrt{3}(E_8 \perp E_8)$ of an order-three automorphism of P_{48p} and give dimensions 42, 41 and 40. Fourth, the dimension-43 packing, which by Corollary 15 cannot come from a bare cross-section at the inherited ambient radius.

Section 5 adds kissing numbers, which Proposition 5 makes a finite count over the ambient's minimal vectors. Corollary 8 of Conway and Sloane [7] constructs twelve cross-section packings of P_{48p} in dimensions 38 to 47 and states their densities, but their kissing numbers appear never to have been computed. We reconstruct all twelve in closed form and count them exactly, and find that in dimensions 42 to 47 their kissing numbers exceed the previously tabulated lower bounds. Our chain does better still in dimensions 40, 41 and 42, and one further cross-section, of P_{48m} and P_{48q}, beats the record in dimension 45 while its density ties the record.

Theorem 1. *Sphere packings exist that give*

$$\delta_{38}^{\text{lat}} \ge \frac{3^{31/2}}{2^{19}\cdot 5} = 9.48\ldots, \quad \delta_{39}^{\text{lat}} \ge \frac{3^{16}}{2^{20}\cdot 5^{1/2}} = 18.35\ldots, \quad \delta_{40}^{\text{lat}} \ge \frac{3^{16}}{2^{20}} = 41.05\ldots,$$

$$\delta_{41}^{\text{lat}} \ge \frac{3^{17}}{2^{21}} = 61.57\ldots, \qquad \delta_{42}^{\text{lat}} \ge \frac{3^{35/2}}{2^{21}} = 106.65\ldots, \qquad \delta_{43} \ge \frac{137^{43/2}}{2^{223/2}\cdot 3^{21}} = 226.51\ldots$$

The first five are attained by lattice packings, that is by bare cross-section packings (3)*, and the last by a union of six translates of a lattice (Table 1).*

Theorem 2. *There are sphere packings with kissing numbers* 1,092,000, 1,324,472, 1,792,386, 7,379,838 *in dimensions* 40, 41, 42 *and* 45 *(Table 2). The first three are lattice packings; the last is a union of four translates of a lattice.*

	previously known		this paper	
dim	construction	kissing number	construction	kissing number
40	binary codes [12, 2, 4]	1,064,368	cross-section Section 3	1,092,000
41	binary codes [12, 2, 4]	1,170,384	cross-section Section 3	1,324,472
42	cross-section [7, Corollary 8]	1,541,292	cross-section Section 3	1,792,386
45	cross-section [7, Corollary 8]	6,702,080	antipode Section 5.3	7,379,838

Table 2: Kissing numbers. The previously known entries are the largest previously tabulated: in dimensions 40 and 41 these are kissing configurations built from binary codes, not known to extend to packings [12]; in dimensions 42 and 45 they are cross-sections of Conway and Sloane [7, Corollary 8], constructed without computing their kissing numbers, which we compute here (Table 5). Our packings in dimensions 40, 41 and 42 are lattices; the dimension-45 one is an antipode packing of $s = 4$ translates of a lattice.

2 Preliminaries

2.1 Cross-sections

Let Λ be an n-dimensional lattice of minimal squared norm μ and let $\mathbb{R}^n = U \oplus V$ be an orthogonal decomposition with $\dim U = k$, $\dim V = l = n - k$. Write

$$K = \Lambda \cap U, \qquad L = \Lambda \cap V, \qquad M = \pi_U(\Lambda),$$

where π_U, π_V are the orthogonal projections, and assume K has rank k. A sublattice $K \subseteq \Lambda$ is *saturated* (or primitive) when Λ/K is torsion-free; $\Lambda \cap U$ is always saturated, and conversely a saturated sublattice is the intersection of Λ with its own span. For Λ unimodular, discriminant gluing identifies the discriminant groups of K and L anti-isometrically [17, Prop. 1.9.8], [7, Thm. 4], so that

$$\det L = \det K = 1/\det M. \tag{2}$$

We write K^* and L^* for the duals of K in U and of L in V. The next fact is used throughout; it needs L saturated, which is automatic, and not K saturated.

Lemma 3. *If Λ is unimodular and $L = \Lambda \cap V$, then $\pi_V(\Lambda) = L^*$. Likewise $\pi_U(\Lambda) = K^*$ when $K = \Lambda \cap U$.*

Proof. For $x \in \Lambda$ and $\ell \in L$ we have $\langle \pi_V(x), \ell \rangle = \langle x, \ell \rangle \in \mathbb{Z}$, so $\pi_V(\Lambda) \subseteq L^*$. Conversely let $y \in L^*$. Since Λ/L is torsion-free it is free, so L is a direct summand of Λ and the homomorphism $\ell \mapsto \langle y, \ell \rangle$ extends to a homomorphism $\Lambda \to \mathbb{Z}$. As Λ is unimodular, that homomorphism is $\langle x, \cdot \rangle$ for some $x \in \Lambda$, and then $\pi_V(x) - y$ is orthogonal to L and lies in $V = \mathrm{span}(L)$, hence vanishes. □

2.2 The antipode construction

Let $S = \{u_0 = 0, u_1, \ldots, u_{s-1}\} \subset M$ have pairwise squared distances at most β for some $\beta < \mu$. The *antipode packing* is

$$\mathcal{A}(S) \;=\; \{\, \pi_V(w) : w \in \Lambda,\ \pi_U(w) \in S \,\},$$

a union of s translates of L packed with spheres of radius $\frac{1}{2}\sqrt{\mu - \beta}$.

Theorem 4 (Conway and Sloane [8])**.** *Drawing spheres of radius $\rho = \frac{1}{2}\sqrt{\mu - \beta}$ around the points of $\mathcal{A}(S)$ gives an l-dimensional sphere packing of center density*

$$\delta(\mathcal{A}(S)) \;=\; \frac{s\,\rho^l}{\sqrt{\det L}}.$$

For unimodular Λ we have $\det L = \det K$ by (2), so this is $s(\mu - \beta)^{l/2} 2^{-l} (\det K)^{-1/2}$, which is (1) when $\mu = 6$. The case $s = 1$, $\beta = 0$ uses the lattice L with spheres of radius $\sqrt{\mu}/2$, the radius inherited from the

ambient minimum. We call this the *bare cross-section packing*; five of our six results are of this kind, with center density

$$\delta_{\mathrm{bare}}(L) \;=\; \frac{(\mu/4)^{l/2}}{\sqrt{\det K}} \;=\; \frac{(3/2)^{(48-k)/2}}{\sqrt{\det K}} \qquad (\mu = 6,\ n = 48). \tag{3}$$

If $\min(L) > \mu$, then L regarded intrinsically as a lattice can of course support larger spheres. Throughout this paper, "bare cross-section packing" means the construction at the inherited ambient radius $\sqrt{\mu}/2$.

The kissing number of an antipode packing is equally explicit: contacts come from minimal vectors of Λ lying in the fibers of π_U over the difference set of S.

Proposition 5. *In the packing $\mathcal{A}(S)$, every sphere in the translate indexed by u_i touches exactly*

$$N_i \;=\; \sum_{\substack{j \neq i \\ |u_i - u_j|^2 = \beta}} \#\{\, w \in \Lambda : |w|^2 = \mu,\ \pi_U(w) = u_j - u_i \,\}$$

other spheres, and the kissing number of $\mathcal{A}(S)$ is $\max_i N_i$.

Proof. Fix a center $\pi_V(w_0)$ with $\pi_U(w_0) = u_i$. Every other center is $\pi_V(w)$ with $\pi_U(w) = u_j$ for some j, and $w \mapsto w - w_0$ is a bijection between such centers and the vectors $v \in \Lambda$ with $\pi_U(v) = u_j - u_i$ (for $j = i$, excluding $v = 0$). The squared distance of the two centers is

$$|\pi_V(w - w_0)|^2 \;=\; |w - w_0|^2 - |u_j - u_i|^2 \;\geq\; \mu - \beta \qquad (w \neq w_0),$$

since $|w - w_0|^2 \geq \mu$ and $|u_j - u_i|^2 \leq \beta$ (for $j = i$ the bound is $\mu > \mu - \beta$). Equality holds if and only if $w - w_0$ is a minimal vector of Λ and $|u_j - u_i|^2 = \beta$; in this case the two spheres of radius $\frac{1}{2}\sqrt{\mu - \beta}$ are tangent. Summing over j gives N_i, which does not depend on the choice of w_0. □

Remark 1. For $s = 1$ the packing $\mathcal{A}(S)$ is the lattice packing of L itself, the contact distance is μ, and the same argument gives $N_0 = \#\{\, w \in \Lambda : |w|^2 = \mu,\ \pi_U(w) = 0 \,\}$: the kissing number of a bare cross-section packing is the number of minimal vectors of Λ that lie in L. Section 5 uses this count.

A convenient sufficient condition for a large cluster, which our dimension-43 packing satisfies, takes S to be the origin together with an entire basis of M:

Theorem 6 (Chen, Hu, Li, Wang and Wu [3])**.** *Let Λ be an n-dimensional unimodular lattice of minimum μ, let K be a saturated cross-section of rank k with Gram matrix $\mathbf{K}$, and put $\mathbf{M} = \mathbf{K}^{-1}$. If $0 < \beta < \mu$ satisfies*

$$\mathbf{M}_{i,i} \leq \beta \qquad \text{and} \qquad \mathbf{M}_{i,i} + \mathbf{M}_{j,j} - 2\mathbf{M}_{i,j} \leq \beta \qquad (1 \leq i < j \leq k),$$

then there is an $(n-k)$-dimensional packing of center density $(k+1)\sqrt{\det \mathbf{M}}\,((\mu - \beta)/4)^{(n-k)/2}$.

Three center densities appear below. Write δ_d for the largest center density of a packing in dimension d, and δ_d^{lat} for the largest center density of a lattice packing, each taken over all packings whether or not one is known; so $\delta_d^{\mathrm{lat}} \leq \delta_d$, and $\delta_d^{\mathrm{lat}} = (\gamma_d/4)^{d/2}$ with γ_d the Hermite constant. A cross-section is a lattice, so what bounds it is δ^{lat}, not δ, and the distinction is not vacuous: among the ranks used below, the densest packing known in dimension 10 is not a lattice [5]. Neither optimum is known in every rank we need, so write $\underline{\delta}_d^{\mathrm{lat}}$ for the largest center density of a lattice packing that is *known*, tabulated by Nebe and Sloane [21]; then $\underline{\delta}_d^{\mathrm{lat}} \leq \delta_d^{\mathrm{lat}}$, with equality exactly when the densest rank-d lattice has been found.

2.3 The ceiling of a bare cross-section

A cross-section of Λ is a lattice of minimum at least $\mu = 6$, and that alone limits how dense its complement can be. The argument is that of Conway and Sloane, who observed that the densest k-dimensional and $(48-k)$-dimensional sections of a unimodular lattice have the same determinant [7, Corollary 5]; with the classification of the densest lattices in dimensions at most 8 [7, Theorem 2] this bounds one in terms of the other. They drew the conclusion in dimensions 43 and above, over even unimodular ambients [7, Corollary 8]; we need it below dimension 43 as well, where their own sections do not attain it (Corollary 11), and over every unimodular ambient.

Proposition 7. *Let K be a cross-section of rank k of a unimodular 48-dimensional lattice of minimum 6, and let γ_k be the Hermite constant. Then*

$$\det K \;\geq\; \left(\frac{6}{\gamma_k}\right)^k,$$

and consequently the bare cross-section packing (3) of its complement has

$$\delta_{\text{bare}} \;\leq\; \delta_k^{\text{lat}}\,(3/2)^{24-k}. \tag{4}$$

Proof. K is a rank-k lattice of minimum $m \geq 6$, so $6/(\det K)^{1/k} \leq m/(\det K)^{1/k} \leq \gamma_k$ by the definition of the Hermite constant, which is the determinant inequality. Substituting it into (3) gives the density. □

The bound depends on neither the choice of 48-dimensional lattice nor extremality: the same argument over an ambient of minimum $\mu \leq 6$ only lowers the right-hand side, and an odd unimodular lattice of rank 48 also has minimum at most 6 by the shadow bound of Rains and Sloane [24], so the ceiling holds over every unimodular ambient of rank 48.

The densest lattice packing has been established in the published literature in every dimension up to 8, so for $k \leq 8$ the bound is effective: $\gamma_5^5 = 8$, $\gamma_6^6 = 64/3$, $\gamma_7^7 = 64$ and $\gamma_8^8 = 256$ [15, 1], attained by D_5, E_6, E_7 and E_8, unique up to similarity [17, Ch. 6].[1] Those are the ranks at which this paper uses published optimality results. Section 4 evaluates the bound rank by rank.

3 A chain of optimal extensions

3.1 The seed: an order-three fixed lattice

Section 2.3 says what to look for in ranks 6, 7, 8: a section isometric to $\sqrt{3}E_6$, $\sqrt{3}E_7$ or $\sqrt{3}E_8$ would attain the ceiling exactly. Symmetry produces all three at once.

Let $\Lambda = P_{48p}$, the extremal even unimodular lattice built from the Pless symmetry code [23, 9], so $\det\Lambda = 1$ and $\mu(\Lambda) = 6$. For an automorphism σ of Λ let $\mathrm{Fix}_\Lambda(\sigma) = \Lambda \cap \ker(\sigma - 1)$.

Proposition 8 (Nebe [18])**.** *There is an automorphism $\sigma \in \mathrm{Aut}(P_{48p})$ of order 3 with $\mathrm{Fix}_{P_{48p}}(\sigma) \cong \sqrt{3}(E_8 \perp E_8)$.*

This is the construction-specific input from the literature: the proof of Theorem 3.8 of Nebe [18] exhibits an order-three σ with $\mathrm{Fix}(\sigma) \cong \sqrt{3}(E_8 \perp E_8)$ in P_{48p}, and Proposition 4.8 there classifies the order-three automorphisms of extremal 48-dimensional lattices. Here $\sqrt{3}R$ means the lattice R with its bilinear form multiplied by 3, so $\det(\sqrt{3}R) = 3^{\operatorname{rank} R}\det R$ and the minimum of $\sqrt{3}R$ is three times that of R.

The two lemmas below are standard, and are proved only to keep the determinants self-contained: invariant lattices are saturated [14, §2] (see [22] for saturated sublattices in general), and a subsystem of simple roots spans a saturated sublattice. It is their conjunction with Proposition 8 that matters, putting $\sqrt{3}E_6$, $\sqrt{3}E_7$ and $\sqrt{3}E_8$ inside an extremal 48-dimensional lattice, where (3) turns them into packings.

Lemma 9. *$F = \mathrm{Fix}_\Lambda(\sigma)$ is saturated in Λ.*

Proof. $F = \Lambda \cap \ker(\sigma - \mathrm{id})$ is the intersection of Λ with a linear subspace, hence saturated by Section 2.1. Directly: if $x \in \Lambda$ and $mx \in F$ for some nonzero integer m, then $m(\sigma x - x) = \sigma(mx) - mx = 0$; a lattice is torsion-free, so $\sigma x = x$ and $x \in F$. □

Lemma 10. *Λ contains saturated sublattices $K_6 \cong \sqrt{3}E_6 \subset K_7 \cong \sqrt{3}E_7 \subset K_8 \cong \sqrt{3}E_8$, of determinants $3^7 = 2187$, $2\cdot 3^7 = 4374$ and $3^8 = 6561$.*

[1] A recent preprint of Dutour Sikirić and van Woerden [11] reports a complete Voronoi enumeration in dimension 9 and concludes that Λ_9 is optimal, equivalently $\gamma_9 = 2$. If this computation is correct, then the $k = 9$ row of Table 4 is also unconditional: $\det K \geq 3^9 = 19683$, and the possible bare-improvement window in dimension 39 is exactly $19683 \leq \det K < 21870$. Because this result is available only as a preprint at the time of writing, we do not use it as an established theorem in the present paper.

Proof. Write $F = F_1 \perp F_2$ with $F_i \cong \sqrt{3}E_8$; each F_i is a direct summand of the free abelian group F. Choose a simple-root basis of F_1 containing the nested Dynkin diagrams $E_6 \subset E_7 \subset E_8$. A subset of a $\mathbb{Z}$-basis spans a saturated subgroup, so the scaled copies of E_6 and E_7 are saturated in F_1. Saturation is transitive, so Lemma 9 gives all three saturated embeddings in Λ. Finally $\det E_6 = 3$, $\det E_7 = 2$ and $\det E_8 = 1$. □

Applying (3) to K_6, K_7, K_8 gives the first three records of Table 1:

$$\delta_{42} \geq \frac{(3/2)^{21}}{\sqrt{3^7}} = \frac{3^{35/2}}{2^{21}}, \qquad \delta_{41} \geq \frac{(3/2)^{41/2}}{\sqrt{2 \cdot 3^7}} = \frac{3^{17}}{2^{21}}, \qquad \delta_{40} \geq \frac{(3/2)^{20}}{\sqrt{3^8}} = \frac{3^{16}}{2^{20}}.$$

Each is exactly the ceiling (4) of its rank, so by Proposition 7:

Corollary 11. *The packings of dimensions* 42, 41 *and* 40 *above are optimal among bare cross-section packings: no bare cross-section packing* (3) *of any unimodular* 48*-dimensional lattice has larger center density in the same dimension. Equivalently, K_6, K_7 and K_8 have the smallest determinant among all cross-sections of rank* 6, 7 *and* 8 *of such lattices, attaining the floor $(6/\gamma_k)^k$ of Proposition 7, and the sections attaining it are unique up to isometry.*

Conway and Sloane claimed optimality only from dimension 43 upwards [7, Corollary 8]. Their sections in dimensions 42, 41 and 40 are $\sqrt{3}D_6$, $\sqrt{3}D_7$ and S_8, of determinants $4 \cdot 3^6$, $4 \cdot 3^7$ and $32 \cdot 3^6$, none of which meets Proposition 7: what was missing was not the bound but a section attaining it. Corollary 11 extends their assertion down to dimension 40.

In the basis of simple roots the three Gram matrices are the leading blocks of

$$3\,C(E_8) \;=\; \begin{bmatrix} 6 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & -3 & 0 & 0 & 0 & 0 \\ -3 & 0 & 6 & -3 & 0 & 0 & 0 & 0 \\ 0 & -3 & -3 & 6 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 6 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 6 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 6 \end{bmatrix},$$

$C(E_8)$ being the Cartan matrix; the leading 6×6 and 7×7 blocks are $3C(E_6)$ and $3C(E_7)$, of determinants 2187 and 4374.

Remark 2. The construction is not specific to P_{48p}. Realizing σ as a signed coordinate permutation of the ternary code frame (for P_{48p} the element $x \mapsto -1/(x+1)$ of $\mathrm{PSL}_2(23)$ acting on the Pless symmetry code, for P_{48q} the corresponding element of $\mathrm{PSL}_2(47)$ on the extended quadratic residue code) and, for P_{48n}, as the product of three generators of the automorphism group listed in the catalogue entry `CQ48a` [20], we obtain the same fixed lattice $\sqrt{3}(E_8 \perp E_8)$, hence the same sections, in three of the four known extremal lattices. Whether the fourth, P_{48m}, has such a section we do not know. An order-three action on $\mathbb{R}^{48}$ with f trivial and t faithful two-dimensional summands has $48 = f + 2t$ and trace $f - t$, so a fixed lattice of rank 16 requires trace 0; an unverified enumeration of order-three elements generated by the catalogue's generators for $\mathrm{Aut}(P_{48m})$ produced only trace -24, that is, only fixed-point-free elements, but we have not certified that computation and nothing in this paper depends on it.

3.2 Extension and the glue radius

We now push past rank 8. The determinant of a one-vector extension is governed by a projection, and integrality restricts that projection to a dual-lattice coset.

Proposition 12. *Let $K = \Lambda \cap U$ be a saturated cross-section of rank k of a unimodular lattice Λ, and $L = \Lambda \cap V$ with $V = U^\perp$.*

1. *Every saturated cross-section of rank $k+1$ containing K is $K + \mathbb{Z}v$ for some $v \in \Lambda$.*
2. $\det(K + \mathbb{Z}v) = \det K \cdot |\pi_V(v)|^2$.

3. *The smallest determinant of a rank-$(k+1)$ cross-section containing K is* $\det K \cdot \mu(L^*)$*, where μ denotes the minimum.*

Proof. (1) If $W \supseteq K$ is a saturated section of rank $k+1$ then W/K is torsion-free of rank 1, hence infinite cyclic; any preimage v of a generator gives $W = K + \mathbb{Z}v$. (2) In a basis of K followed by v, the Gram matrix of $K + \mathbb{Z}v$ has the Gram matrix of K as its leading block and $|v|^2$ in the corner; its Schur complement is $|v|^2 - |\pi_U(v)|^2 = |\pi_V(v)|^2$. (3) By Lemma 3, π_V maps Λ onto L^*, so the values $|\pi_V(v)|^2$ for $v \in \Lambda$ with $\pi_V(v) \neq 0$ are exactly the nonzero squared norms of L^*, and the smallest is $\mu(L^*)$. If the resulting $K + \mathbb{Z}v$ is not saturated, its saturation is a section of the same rank and strictly smaller determinant containing K, so the minimum is attained by a saturated section. □

Parts (2) and (3) are the cross-section form of a computation of Conway and Sloane. Writing Λ_n as a union of layers of a sublattice Λ_{n-1}, their equation (2) reads $X_n = \gamma_{n-1} X_{n-1}$, with X a determinant and γ_{n-1} the minimum of the projection of Λ_n orthogonal to Λ_{n-1} [7, §II]. They grow a lattice; we grow a section inside a fixed ambient, where Lemma 3 identifies the projection as L^*.

Part (3) replaces a search through Λ by one shortest-vector computation, but in the complement, of dimension 39 or more. The next lemma moves the question into rank k, where a finite enumeration settles it, and removes the dependence on the ambient.

Lemma 13. *Let Λ be an integral lattice of minimum μ, let $K \subseteq \Lambda$ have rank k and span U, put $V = U^\perp$, and set*

$$\rho_{\mathrm{glue}}(K) \;=\; \max_{y \in K^*} \operatorname{dist}(y, K),$$

a maximum over the finitely many classes of K^*/K. We call $\rho_{\mathrm{glue}}(K)$ the glue radius*: it is the largest norm of a minimal glue vector of K [9, Ch. 4]. Then every $v \in \Lambda$ with $\pi_V(v) \neq 0$ satisfies*

$$|\pi_V(v)|^2 \;\geq\; \mu - \rho^2_{\mathrm{glue}}(K).$$

Moreover $\rho_{\mathrm{glue}}(K) \leq \rho_{\mathrm{cov}}(K)$, where $\rho_{\mathrm{cov}}(K)$ is the ordinary covering radius of K.

Proof. Since Λ is integral, $\langle v, x \rangle \in \mathbb{Z}$ for all $x \in K$, so $\pi_U(v) \in K^*$. Choose $x \in K$ nearest to $\pi_U(v)$; then $|\pi_U(v) - x|^2 \leq \rho^2_{\mathrm{glue}}(K)$. The vector $v - x$ lies in Λ and is nonzero because $\pi_V(v - x) = \pi_V(v) \neq 0$, so

$$\mu \;\leq\; |v - x|^2 \;=\; |\pi_V(v)|^2 + |\pi_U(v) - x|^2 \;\leq\; |\pi_V(v)|^2 + \rho^2_{\mathrm{glue}}(K).$$

Finally, the ordinary covering radius maximizes $\operatorname{dist}(y, K)$ over all $y \in U$, while the glue radius maximizes it only over $y \in K^*$, hence $\rho_{\mathrm{glue}}(K) \leq \rho_{\mathrm{cov}}(K)$. □

The argument is again theirs: equation (1) of [7, §II] reads $\gamma_{n-1} \geq 4 - h^2_{n-1}$, with h_{n-1} the distance from the extension vector to the layer lattice, a quantity they call the *subcovering radius* and then bound by the ordinary covering radius. In the present integral setting, $\pi_U(v)$ is restricted to K^* rather than being an arbitrary point of U. Thus the intrinsic glue radius gives a bound no weaker than the ordinary covering-radius bound and is computable by a finite enumeration of K^*/K. The difference decides the question at hand. The ordinary covering radius of $\sqrt{3}E_8$ is $\sqrt{3}$, so bounding by it gives only $\det \geq 3^8(6-3) = 19683$, which leaves dimension 39 open, whereas $\rho^2_{\mathrm{glue}}(\sqrt{3}E_8) = 8/3$ gives $3^8(6-8/3) = 21870$, and 21870 is attained.

3.3 The chain

Combining Proposition 12(2) with Lemma 13: every rank-$(k+1)$ cross-section containing K has determinant at least $\det K\,(\mu - \rho^2_{\mathrm{glue}}(K))$, in any integral ambient of minimum μ. Table 3 evaluates ρ^2_{glue} for the sections at hand. Each row's bound is *attained* by the next row's section, so each step of the chain is optimal.

Theorem 14. *Let Λ be a unimodular 48-dimensional lattice of minimum 6 containing a saturated $\sqrt{3}E_8$. Then:*

1. *every rank-9 cross-section of Λ containing that $\sqrt{3}E_8$ has determinant at least 21870;*
2. *in P_{48p}, P_{48q} and P_{48n} the value 21870 is attained by a saturated section K_9, whose complement is a 39-dimensional lattice packing of center density $2^{-20}3^{16}5^{-1/2} = 18.35\ldots$;*

K	$\det K$	$\rho^2_{\mathrm{glue}}(K)$	$6-\rho^2_{\mathrm{glue}}(K)$	$\det K\,(6-\rho^2_{\mathrm{glue}}(K))$	attained by	complement
$\sqrt{3}E_6$	2187	4	2	4374	$\sqrt{3}E_7$	dim 41, 61.57…
$\sqrt{3}E_7$	4374	9/2	3/2	6561	$\sqrt{3}E_8$	dim 40, 41.05…
$\sqrt{3}E_8$	6561	8/3	10/3	21870	K_9	dim 39, 18.35…
K_9	21870	7/2	5/2	54675	K_{10}	dim 38, 9.48…
K_{10}	54675	34/9	20/9	121500		dim 37, $\le 5.19\ldots$

Table 3: The chain of optimal extensions. By Lemma 13 every rank-$(k+1)$ cross-section containing K has determinant at least $\det K\,(6-\rho^2_{\mathrm{glue}}(K))$, in any integral ambient of minimum 6, where $k=\operatorname{rank} K$; the fifth column is this floor, and the sixth shows that each bound is attained by the next section of the chain, so every step is optimal. The values of ρ^2_{glue} are exact, obtained by enumerating all $\det K$ classes of K^*/K; the maximum is attained by 2, 1, 1920, 561 and 324 classes respectively.

3. *every rank-*10 *cross-section containing that* K_9 *has determinant at least* 54675*, attained by a saturated* K_{10} *whose complement is a* 38*-dimensional lattice packing of center density* $2^{-19}3^{31/2}5^{-1}=9.48\ldots$*;*

4. *every rank-*11 *cross-section containing* K_{10} *has determinant at least* 121500*, so its complement has center density at most* $5.19\ldots$*, below the record in dimension* 37*.*

Proof. The lower bounds are Lemma 13 with the values of ρ^2_{glue} in Table 3, computed exactly by enumerating the classes of K^*/K (Section 6); the attainments are the certified sections of Appendix A, and the densities are (3). For (4), $(3/2)^{37/2}/\sqrt{121500}=5.19\ldots$ while the dimension-37 record is $2^{5/2}=5.65\ldots$. □

The Gram matrices of the two new sections, in the bases of Appendix A, are

$$\mathbf{K}_9 = \begin{bmatrix} 6 & -3 & -3 & -3 & -3 & -3 & -3 & 3 & -2 \\ -3 & 6 & 3 & 3 & 3 & 3 & 3 & 0 & 1 \\ -3 & 3 & 6 & 0 & 3 & 0 & 0 & -3 & 3 \\ -3 & 3 & 0 & 6 & 3 & 3 & 3 & 0 & -1 \\ -3 & 3 & 3 & 3 & 6 & 3 & 3 & -3 & 1 \\ -3 & 3 & 0 & 3 & 3 & 6 & 3 & 0 & 0 \\ -3 & 3 & 0 & 3 & 3 & 3 & 6 & 0 & 1 \\ 3 & 0 & -3 & 0 & -3 & 0 & 0 & 6 & -1 \\ -2 & 1 & 3 & -1 & 1 & 0 & 1 & -1 & 6 \end{bmatrix},$$

$$\mathbf{K}_{10} = \begin{bmatrix} 6 & -3 & -3 & -3 & -3 & -3 & -3 & 3 & -2 & 1 \\ -3 & 6 & 3 & 3 & 3 & 3 & 3 & 0 & 1 & -1 \\ -3 & 3 & 6 & 0 & 3 & 0 & 0 & -3 & 3 & 1 \\ -3 & 3 & 0 & 6 & 3 & 3 & 3 & 0 & -1 & 0 \\ -3 & 3 & 3 & 3 & 6 & 3 & 3 & -3 & 1 & -1 \\ -3 & 3 & 0 & 3 & 3 & 6 & 3 & 0 & 0 & -2 \\ -3 & 3 & 0 & 3 & 3 & 3 & 6 & 0 & 1 & -2 \\ 3 & 0 & -3 & 0 & -3 & 0 & 0 & 6 & -1 & 1 \\ -2 & 1 & 3 & -1 & 1 & 0 & 1 & -1 & 6 & 2 \\ 1 & -1 & 1 & 0 & -1 & -2 & -2 & 1 & 2 & 6 \end{bmatrix},$$

of determinants $21870 = 2\cdot 3^7\cdot 5$ and $54675 = 3^7\cdot 5^2$. The two bases are nested: the leading 8×8 block of each is a Gram matrix of $\sqrt{3}E_8$, and $\mathbf{K}_9$ is the leading 9×9 block of $\mathbf{K}_{10}$, so the inclusions $\sqrt{3}E_8 \subset K_9 \subset K_{10}$ of Theorem 14 can be read off the matrices rather than taken on trust. Neither is a rescaled root lattice, since $21870/3^9$ and $54675/3^{10}$ are not integers; their discriminant groups are $(\mathbb{Z}/3)^6\times\mathbb{Z}/30$ and $(\mathbb{Z}/3)^4\times\mathbb{Z}/15\times\mathbb{Z}/45$, and they have 258 and 294 minimal vectors.

Remark 3. Concretely, since K_8 is saturated, Lemma 3 gives $\pi_U(\Lambda)=K_8^*=\frac{1}{\sqrt{3}}E_8$, so every $v\in\Lambda$ has $\pi_U(v)=e/\sqrt{3}$ with $e\in E_8$, and $|\pi_V(v)|^2=|v|^2-|e|^2/3$: a *larger* E_8-component means a *smaller* determinant. For a minimal vector v and every root ρ of E_8 we have $|v-\sqrt{3}\rho|^2\ge 6$, i.e. $(e,\rho)\le 3$, and an enumeration of

the vectors of E_8 of norm at most $|e|^2 \le 3|v|^2 = 18$ shows that this forces $|e|^2 \le 8$. The bound is attained: of the 52,416,000 minimal vectors of P_{48p}, exactly 272,160 have $|e|^2 = 8$, and each of them extends K_8 to a section of determinant $3^8(6 - 8/3) = 21870$. In the Leech lattice the same computation gives only $(e, \rho) \le 2$ and no comparable gain, which is why this route is special to minimum 6.

Remark 4. For K_9 and K_{10}, our exhaustive cluster maximization over the duals, with no cap on the number of dual points or on the clique search and with $\beta \le 1$, returns $s = 1$. Thus no cluster with $\beta \le 1$ improves the bare packing. We do not infer anything here about clusters of diameter $\beta > 1$; such clusters may contain additional dual points and are not ruled out by this computation.

4 Dimension 43: beyond the bare cross-section

Evaluating Proposition 7 rank by rank says how far the bare route reaches. In rank 5 it stops short of dimension 43.

Corollary 15. *No bare cross-section of any unimodular* 48*-dimensional lattice gives a* 43*-dimensional packing of center density above* $2^{-45/2}3^{19} = 195.94\ldots$*, and that value is attained by the section* $\sqrt{3}D_5$ *of determinant* 972*. In particular the* 43*-dimensional record of Table 1 cannot be obtained from a bare cross-section. This is the second assertion of Corollary 8 of Conway and Sloane [7], stated there for every even unimodular* 48*-dimensional lattice of minimum* 6 *and for all dimensions from* 43 *upwards.*

Proof. Proposition 7 with $k = 5$ and $\gamma_5^5 = 8$ gives $\det K \ge 6^5/8 = 972$ and $\delta_{\text{bare}} \le (3/2)^{43/2}/\sqrt{972} = 2^{-45/2}3^{19}$, with equality only for $\sqrt{3}D_5$, which is the dimension-43 section of Conway and Sloane [7, Corollary 8]. □

k	$48-k$	densest known lattice	bound / benchmark	status
5	43	D_5	$195.94\ldots$	known 43-dim. packing $226.51\ldots$ exceeds this ceiling: no bare cross-section reaches it
6	42	E_6	$106.65\ldots$	attained exactly; the section is optimal
7	41	E_7	$61.57\ldots$	attained exactly; the section is optimal
8	40	E_8	$41.05\ldots$	attained exactly; the section is optimal
9	39	Λ_9	$19.35\ldots$	bare improvement possible; $\det K < 19683$ would also improve the rank-9 lattice record
10	38	Λ_{10}	$10.53\ldots$	bare improvement possible; $\det K \le 44286$ would also improve the rank-10 lattice record

Table 4: Bare cross-section bounds and current-record benchmarks. For $k \le 8$ (above the rule), the displayed value is the unconditional ceiling of Proposition 7, because the Hermite constant is known from published results. For $k = 9, 10$ we instead display the benchmark $\underline{\delta}_k^{\text{lat}}(3/2)^{24-k}$ from the densest lattice currently known in dimension k [21]. Exceeding this benchmark with a bare cross-section would simultaneously produce a new k-dimensional lattice record. A recent preprint, discussed in Section 2.3, claims that the $k = 9$ benchmark is in fact unconditional.

For $k = 9, 10$ we do not use an established value of the Hermite constant, so putting the current record $\underline{\delta}_k^{\text{lat}}$ in place of δ_k^{lat} gives only a benchmark. Since $\underline{\delta}_k^{\text{lat}} \le \delta_k^{\text{lat}}$, the resulting determinant benchmark $(3/2)^k/(\underline{\delta}_k^{\text{lat}})^2$ is at least as large as the true determinant floor, whereas the density benchmark $\underline{\delta}_k^{\text{lat}}(3/2)^{24-k}$ is at most as large as the true ceiling. Thus crossing either benchmark would establish a new rank-k lattice record, but failing to cross it does not rule out a bare improvement in dimensions $48 - k$. Concretely, in dimension 39 a section with $19683 \le \det K < 21870$ would improve our packing without necessarily improving the current rank-9 lattice record; a smaller determinant would improve both. In dimension 38, a section with

$44287 \le \det K < 54675$ would improve our packing without necessarily improving the current rank-10 lattice record, whereas $\det K \le 44286$ would improve both. The exact rank-10 benchmark is $3^{11}/4 = 44286.75$, so the last inequality is equivalent because $\det K$ is integral.

No bare cross-section, then, beats $195.94\ldots$ in dimension 43. A cluster does, and this is where the second lever in (1) earns its keep.

Let K be the 5-dimensional lattice with Gram matrix

$$\mathbf{K} = \begin{bmatrix} 6 & -1 & -1 & -2 & -1 \\ -1 & 6 & -2 & -1 & -1 \\ -1 & -2 & 6 & -1 & -1 \\ -2 & -1 & -1 & 6 & -1 \\ -1 & -1 & -1 & -1 & 6 \end{bmatrix}, \qquad \mathbf{M} = \mathbf{K}^{-1} = \frac{1}{48} \begin{bmatrix} 14 & 7 & 7 & 8 & 6 \\ 7 & 14 & 8 & 7 & 6 \\ 7 & 8 & 14 & 7 & 6 \\ 8 & 7 & 7 & 14 & 6 \\ 6 & 6 & 6 & 6 & 12 \end{bmatrix},$$

so $\det \mathbf{K} = 3072 = 2^{10} \cdot 3$. It is a saturated cross-section of P_{48n} and of P_{48m} (Appendix A). The hypotheses of Theorem 6 hold with $\beta = 7/24$: the diagonal entries of $\mathbf{M}$ are $14/48$ and $12/48$, and the quantities $\mathbf{M}_{i,i} + \mathbf{M}_{j,j} - 2\mathbf{M}_{i,j}$ are all $12/48$ or $14/48$. The cluster is $S = \{0, m_1, \ldots, m_5\}$, the origin together with the basis of M dual to the chosen basis of K; its six points form three pairs at squared distance $1/4$, namely $\{0, m_5\}$, $\{m_1, m_4\}$ and $\{m_2, m_3\}$, and all other distances equal $\beta = 7/24$. Hence

$$\delta \;=\; 6 \cdot \frac{1}{\sqrt{3072}} \left(\frac{6 - \frac{7}{24}}{4} \right)^{43/2} \;=\; \frac{137^{43/2}}{2^{223/2} \cdot 3^{21}} \;=\; 226.51\ldots,$$

improving the 1982 record $195.94\ldots$ by a factor $1.15\ldots$ that splits as $6 \cdot \sqrt{972/3072} \cdot (137/144)^{43/2}$: cluster size, determinant penalty, radius penalty. The lattice K is not a rescaled root lattice ($3072/3^5 \notin \mathbb{Z}$), has 12 minimal vectors, and has discriminant group $\mathbb{Z}/4 \times \mathbb{Z}/16 \times \mathbb{Z}/48$.

5 Kissing numbers

Proposition 5 turns a kissing number into a finite count over the complete list of 52,416,000 minimal vectors of the ambient, so every packing in this paper comes with one for free. Three families are worth recording: the 1982 cross-sections, whose kissing numbers appear never to have been computed; the chain of Section 3, which improves the tabulated bounds in dimensions 40, 41 and 42; and one further packing in dimension 45. Table 6 collects what the three give in dimensions 40 to 47.

5.1 The 1982 cross-sections

Corollary 8 of Conway and Sloane [7] constructs cross-sections of P_{48p} orthogonal to

$$\sqrt{3}A_1,\ \sqrt{3}A_2,\ \sqrt{3}A_3,\ \sqrt{3}D_3,\ \sqrt{3}D_4,\ \ldots,\ \sqrt{3}D_7,\ S_7,\ S_8,\ S_9,\ S_{10},$$

where $S_n = \langle \sqrt{3}D_{n-1}, v^* \rangle$ are the glue lattices of the corollary's proof. The resulting bare packings live in dimensions 47 down to 38, with two in dimension 41 ("D_7 or S_7" in the corollary) and, as the table shows, two in dimension 45 as well. The corollary states their densities but not their kissing numbers. We reconstruct all twelve in closed form, in the coordinate frame of the corollary's own proof, and count exactly (Table 5).

The count needs an explicit embedding, and the proof supplies one: it fixes coordinates, taken from Leech and Sloane [16], in which the minimal vectors of P_{48p} include all vectors with two entries ± 3 and the rest zero, together with $v^* = (-2, 1^{14}, 0^{33})$. Our frame for P_{48p} is exactly that frame, so those vectors exist verbatim and a vector of the shape of v^* exists on the support of every weight-15 codeword. We pin the lexicographically first such codeword and write every complement basis down in closed form, with no search. Each kissing number is then one sweep of the minimal shell, counting the fiber over zero (Remark 1).

In dimensions 42 to 47 these counts exceed the previously tabulated lower bounds; in dimensions 38 to 41 they fall below them. The packings that held the density records for four decades were also hiding the best kissing numbers, in the range where nobody had looked.

dim	complement	kissing number	previously tabulated
47	$\sqrt{3}A_1$	**23,766,960**	9,741,412
46	$\sqrt{3}A_2$	**12,309,600**	5,318,060
45	$\sqrt{3}A_3$	6,687,648	3,047,160
45	$\sqrt{3}D_3$	**6,702,080**	3,047,160
44	$\sqrt{3}D_4$	**4,153,600**	2,948,552
43	$\sqrt{3}D_5$	**2,545,056**	1,745,692
42	$\sqrt{3}D_6$	**1,541,292**	1,250,676
41	$\sqrt{3}D_7$	922,282	1,170,384
41	S_7	922,664	1,170,384
40	S_8	578,368	1,064,368
39	S_9	360,716	755,988
38	S_{10}	224,608	566,652

Table 5: The twelve cross-section packings of Conway and Sloane [7, Corollary 8] and their kissing numbers, computed here. Bold entries exceed the previously tabulated lower bounds [12, 2, 4]. The two dimension-45 rows are two embeddings of one lattice ($A_3 = D_3$), which the corollary's family contains both as the A-series and as the D-fork; see Remark 5.

Remark 5. Unlike the density, which by Theorem 4 depends only on $\det L$, the kissing number of a cross-section packing depends on the embedding of the complement in the ambient lattice. The two dimension-45 rows of Table 5 show this concretely: $\sqrt{3}A_3$ and $\sqrt{3}D_3$ are the same lattice, but the corollary's family contains it twice, once as the A-series and once as the D-fork, and the two embeddings give 6,687,648 and 6,702,080 contacts. Since the corollary pins the embeddings only up to its frame condition, the values in Table 5 are those of the canonical frame fixed above.

5.2 The chain

The complements of K_6, K_7 and K_8 improve on Table 5 in their own dimensions, and on the tabulated bounds as well:

$$1{,}792{,}386, \qquad 1{,}324{,}472, \qquad 1{,}092{,}000$$

in dimensions 42, 41 and 40. Only the first competes with a 1982 section: in dimension 42 the bound to beat is Table 5's 1,541,292, while in dimensions 41 and 40 the 1982 sections fall well short of the tabulated 1,170,384 and 1,064,368, which are therefore the relevant comparison. All three counts are the same in P_{48p}, P_{48q} and P_{48n}.

Remark 6. The last two sections of the chain do not improve their dimensions. The complement of K_9 has 604,644, 604,472 and 604,892 contacts in P_{48p}, P_{48q} and P_{48n}, and that of K_{10} has 381,658, 381,438 and 381,620, all below the records 755,988 and 566,652. That the three counts differ is itself informative: by Theorem 4 the three embeddings of each section give packings of equal density, so they are inequivalent under the automorphism groups of their ambients even though no density computation can distinguish them.

5.3 A kissing record in dimension 45

Ranking cross-sections by the kissing number of their antipode packing rather than by density yields one more packing, which beats even the corresponding 1982 cross-section. Here K is the 3-dimensional cross-section of P_{48m} with Gram matrix

$$\mathbf{K} = \begin{bmatrix} 6 & -2 & -2 \\ -2 & 6 & -2 \\ -2 & -2 & 6 \end{bmatrix}, \qquad \mathbf{M} = \mathbf{K}^{-1} = \frac{1}{8}\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad \det \mathbf{K} = 128,$$

isometric to $2\sqrt{2}A_3^*$, the rescaled body-centered cubic lattice. The hypotheses of Theorem 6 hold with $\beta = \frac{1}{4}$, all pairwise squared distances of $S = \{0, m_1, m_2, m_3\}$ being exactly $\frac{1}{4}$, and the center density

$$\delta \;=\; 4 \cdot \frac{1}{\sqrt{128}} \left(\frac{6 - \frac{1}{4}}{4} \right)^{45/2} \;=\; \frac{23^{45/2}}{2^{183/2}} \;=\; 1243.45\ldots$$

equals the record of Chen, Hu, Li, Wang and Wu [3]. Every fiber of Proposition 5 contains 2,459,946 minimal vectors and the per-sphere counts are equal across translates, so every sphere touches

$$3 \cdot 2{,}459{,}946 \;=\; 7{,}379{,}838$$

others, against 6,702,080 for the better of the two 1982 dimension-45 cross-sections. The section embeds in P_{48q} as well, giving a packing of the same density by Theorem 4 and, although the kissing number need not survive a change of embedding (Remark 5), an exact count there returns the same 7,379,838.

dim	previously tabulated	CS82 [7, Cor 8]	this paper
40	1,064,368	578,368	**1,092,000**
41	1,170,384	922,664	**1,324,472**
42	1,250,676	1,541,292	**1,792,386**
43	1,745,692	**2,545,056**	
44	2,948,552	**4,153,600**	
45	3,047,160	6,702,080	**7,379,838**
46	5,318,060	**12,309,600**	
47	9,741,412	**23,766,960**	

Table 6: Kissing number lower bounds in dimensions 40 to 47. The middle column is Table 5, the kissing numbers of the 1982 cross-sections; the last is this paper's chain (dimensions 40 to 42) and the dimension-45 packing of Section 5.3. Bold marks the largest entry of each row, and every one of them improves on the previously tabulated bound [12, 2, 4]. For comparison the best known upper bounds in dimensions 44 to 47 are 220,788,272, 316,735,249, 441,900,184 and 621,658,419 [10, 4].

6 Search and verification

Everything that decides a claim in this paper is computed in exact integer or rational arithmetic; floating point and lattice reduction appear only inside search heuristics.

The seed of Section 3.1 needs no search: σ is written down in closed form on the coordinate frame of the ternary code, its fixed lattice is an integer kernel computation, and the 480 minimal vectors of that fixed lattice split into two orthogonal root systems of type E_8, from which a basis with Gram matrix $3C(E_8)$ is read off.

The chain of Section 3 was found by sweeping the complete list of 52,416,000 minimal vectors of the ambient once, classifying each by its projection onto the span of the current section (Remark 3), and ranking extensions by the exact Schur complement of Proposition 12(2). The optimality of each step is not a search result: $\rho^2_{\mathrm{glue}}(K)$ is computed by enumerating the $\det K$ classes of K^*/K (2187, 4374, 6561, 21870 and 54675 classes respectively) and taking, for each class, the minimum norm of its representatives, a finite computation of rank at most 10. As a check, the dual shortest-vector problem of Proposition 12(3) was also solved directly in the complement for the step $\sqrt{3}E_8 \to K_9$, with the enumeration running to completion, and returned the same value $\mu(L^*) = 10/3$.

The dimension-43 section of Section 4 was found by simulated annealing over 5-tuples of ambient lattice vectors: a move replaces or perturbs one vector of the tuple, each tuple is completed to the saturated section its span meets, and the section is scored by an exact lower bound for the best antipode density over its dual; the best candidates are then re-scored exhaustively, enumerating all short dual points and running a maximum-clique search over admissible pairs for each realizable diameter. Independent runs over P_{48n} and over P_{48m} converged to isometric sections.

Each packing is recorded as a machine-checkable certificate that names the ambient lattice, the section basis in ambient coordinates and the cluster, and a verifier rebuilds the ambient from its published construction and re-proves membership, saturation, $\det L = \det K$, the cluster distances, the minimum, and the density. The code, the verifier and certificates for every packing in this paper, in each of the ambients where it exists, are available at `https://github.com/wcgbg/sphere-packing`.

7 Open problems

Table 4 localizes what is left for the bare cross-section route and separates unconditional ceilings from current-record benchmarks. In dimensions 40, 41 and 42 the bare route is closed: the sections of Corollary 11 are optimal among bare cross-section packings. This does not rule out antipode clusters over these or other sections. In dimension 43 the bare route is also closed, while the cluster route is essential for our improvement and may admit further gains.

The interesting bare cases are dimensions 39 and 38. In dimension 39, any rank-9 section with determinant below 21870 would improve our present packing. The current published rank-9 lattice record corresponds to the benchmark 19683: under the published-literature convention used here, a determinant below that is not excluded by an optimality theorem, but would simultaneously produce a new 9-dimensional lattice record. Thus $[19683, 21870)$ is the portion of the search interval that could improve dimension 39 without necessarily improving that lower-dimensional record. If the recent dimension-9 preprint cited above is correct, then determinants below 19683 are excluded and this interval is the full rigorous search window. Likewise, for dimension 38 one seeks determinant below 54675. The current rank-10 lattice record gives the exact benchmark $3^{11}/4 = 44286.75$; because section determinants are integral, $\det K \le 44286$ would also improve the rank-10 lattice record, while $44287 \le \det K < 54675$ could improve dimension 38 without doing so.

Any such section inside an even ambient lattice must itself be an even integral lattice of minimum at least 6, which is a real constraint: at the rank-9 benchmark $19683 = 3^9$ the natural candidate, the rescaled laminated lattice $\sqrt{3/2}\,\Lambda_9$, is not integral, so a rank-9 section of that determinant would have to be some other lattice. We do not know which even integral lattices of minimum 6 occur in ranks 9 and 10, and determining their smallest determinant would sharpen these searches. The current lattice records alone do not provide rigorous determinant floors in these ranks; the recent dimension-9 preprint, if correct, supplies the rank-9 floor. We leave the corresponding integral-section questions open.

AI Disclosure

We used Claude and ChatGPT to assist with background research, generating and exploring ideas, developing the search and verification code, checking derivations, running and analyzing the computations, and drafting and editing this paper. All computational claims underlying the new packings and kissing-number computations are independently established by the exact-arithmetic certificates and verifier of Section 6. The theoretical and literature-based claims are supported by the arguments and references in the paper. The authors have reviewed all content and take full responsibility for it.

A Certificates

Each packing is specified by integer matrices: $\mathbf{G}$, the 48×48 Gram matrix of a basis $b_1, \dots, b_{48}$ of the ambient lattice Λ; $\mathbf{T}$, a $k \times 48$ matrix whose row i holds the coordinates of the i-th basis vector of the cross-section with respect to $b_1, \dots, b_{48}$; and, when the cluster is not trivial, $\mathbf{W}$, an $s \times 48$ matrix whose row i holds the coordinates of a lattice point w_i with $\pi_U(w_i) = u_i$ (row 0 being $w_0 = 0$). Identifying Λ with $\mathbb{Z}^{48}$ under $\langle x, y\rangle = x\mathbf{G}y^{\mathsf{T}}$, the following certificate-local claims can be checked in exact arithmetic:

1. $\mathbf{G}$ is symmetric and positive definite with even diagonal and $\det \mathbf{G} = 1$, so Λ is an even unimodular lattice of rank 48; positive definiteness can be checked in exact arithmetic, for example by an exact LDL^{T} decomposition;

2. $\mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}} = \mathbf{K}$, the Gram matrix displayed in the text, with the stated determinant;

3. the greatest common divisor of the $k \times k$ minors of $\mathbf{T}$ is 1 (equivalently, the Smith normal form of $\mathbf{T}$ is $[\, I_k \mid 0 \,]$), so the rows of $\mathbf{T}$ span a saturated sublattice: $K = \Lambda \cap U$ exactly;

4. $\mathbf{W}\mathbf{G}\mathbf{T}^{\mathsf{T}}$ is a zero row above I_k, which is the statement $S = \{0, m_1, \ldots, m_k\}$ of Theorem 6;

5. the pairwise squared distances $d^2(u_i, u_j) = (c_i - c_j)\mathbf{K}^{-1}(c_i - c_j)^{\mathsf{T}}$ are as listed, all at most $\beta < 6$, with β attained;

6. Theorem 4 with $\det L = \det \mathbf{K}$ gives the stated density.

The packing itself is determined by the same data: $L = \{x \in \mathbb{Z}^{48} : x\mathbf{G}\mathbf{T}^{\mathsf{T}} = 0\}$ is an integer kernel, and $\mathcal{A}(S) = \bigcup_i (w_i - \pi_U(w_i) + L)$.

One claim is not certified by the printed data: the minimum $\mu = 6$ of the ambient. For P_{48p}, $\mathbf{G}$ is the Gram matrix of the basis built from the Pless symmetry code [23, 9]; for P_{48m} it is the catalogue entry `P48m`. In both cases $\mu = 6$ is the published extremality of the ambient, and a reader who prefers not to rely on it can re-verify $\mu = 6$ from $\mathbf{G}$ by a short-vector enumeration. Certificates for the same packings in the other ambients (P_{48q} and P_{48n} for dimensions 38–42, P_{48n} for dimension 43, P_{48q} for dimension 45) are in the repository.

A.1 Dimensions 42, 41 and 40

Certificates
`records/dim42-delta106/P48p-e6f899b4`,
`records/dim41-delta61/P48p-28a12f16`, and
`records/dim40-delta41/P48p-6a1733b0`.

The three sections are the row prefixes of one matrix: rows 1–6 span $K_6 \cong \sqrt{3}E_6$, rows 1–7 span $K_7 \cong \sqrt{3}E_7$, and all eight rows span $K_8 \cong \sqrt{3}E_8$, with $\mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}} = 3\,C(E_8)$.

G (the Gram matrix of P_{48p}):

```
[[6 3 3 -3 3 -3 2 -3 -2 -3 3 3 -1 1 -1 -3 -2 -1 -2 -2 0 1 2 -1 0 0 2 2 -1 0 -2 -1 -1 0 -2 1 -2 1 -1 -2 0 3 -3 -3 -1 0 2 -2]
 [3 6 3 0 3 -3 0 -1 1 -3 3 3 1 0 0 0 -3 -2 -2 -2 0 3 1 2 -2 -1 2 0 0 -1 0 1 0 -1 0 0 -1 -2 1 2 2 -1 -2 -2 -2 0 -1]
 [3 3 6 -3 3 -3 -1 -3 0 -3 3 3 2 -1 -2 0 -3 -2 -3 -2 0 2 3 2 0 1 0 0 1 2 -1 1 1 -2 1 -1 0 0 -1 -1 1 3 -3 -3 1 0 1 0]
 [-3 0 -3 6 0 0 1 2 0 2 0 0 1 -1 3 1 1 0 1 0 0 -3 -1 -1 0 -2 -2 1 -1 -2 1 -1 1 0 1 -1 1 -2 1 1 1 -2 2 2 -2 -1 -2 2]
 [3 3 3 0 6 -3 2 -3 -2 -3 3 3 1 -2 1 0 -1 -3 -2 -3 1 0 1 1 -1 0 -1 2 0 -1 0 0 0 0 1 -1 0 -2 -2 -2 1 3 -1 -2 0 -2 1 0]
 [-3 -3 -3 0 -3 6 0 1 2 1 -3 -3 0 -1 0 1 2 0 2 3 0 1 -2 -1 -1 0 1 -1 2 0 0 -1 1 2 1 2 1 -1 0 1 -2 -2 2 2 2 2 1 -1]
 [2 0 -1 1 2 0 6 0 -1 0 2 1 0 -1 0 -1 1 -2 1 -2 0 -2 -2 -2 -2 0 2 1 -2 -1 -1 -3 1 2 -1 1 -2 -1 1 -1 -2 1 0 0 0 0 0 -2]
 [-3 -1 -3 2 -3 1 0 6 1 3 -2 -3 -1 2 1 1 0 2 1 0 -1 -1 -1 0 2 0 0 0 -1 1 0 1 0 0 -1 -1 -1 1 1 2 -1 -1 3 1 0 0 -3 0]
 [-2 1 0 0 -2 2 -1 1 6 -1 0 -1 2 0 -1 1 1 -1 2 0 -2 1 1 1 1 0 -1 -2 1 1 0 0 3 1 0 0 0 -1 0 2 1 -1 0 2 1 -1 0 0]
 [-3 -3 -3 2 -3 1 0 3 -1 6 -3 -3 -2 0 1 2 1 3 2 2 -1 -2 -3 -1 -1 0 1 0 -2 1 2 0 0 -1 0 -1 1 0 2 2 -2 -3 2 1 -1 2 -2 0]
 [3 3 3 0 3 -3 2 -2 0 -3 6 3 1 -1 0 -1 -1 -2 -2 -3 1 -1 2 1 0 -1 0 1 -1 0 -2 -2 1 -1 0 0 -2 0 0 -1 2 3 -2 -1 0 -1 1 -1]
 [3 3 3 0 3 -3 1 -3 -1 -3 3 6 1 0 0 -2 -1 -2 -2 -1 1 0 2 -1 -1 -1 0 0 1 -1 -1 -1 0 -1 0 0 -1 0 -1 -2 2 1 -2 -2 -1 -1 1 1]
 [-1 1 2 1 1 0 0 -1 2 -2 1 1 6 -1 -2 1 -1 -3 0 -1 0 0 1 1 0 0 -1 -1 1 -1 -1 1 2 0 1 -1 1 -2 1 0 1 0 -1 1 0 0 0 1]
 [1 0 -1 -1 -2 -1 -1 2 0 0 -1 0 -1 6 -1 -2 -1 3 1 1 0 1 1 -2 1 1 1 -1 0 -1 -2 2 -2 -1 -2 -1 -3 2 1 -1 0 0 -1 -1 -2 0 -1 0]
 [-1 0 -2 3 1 0 0 1 -1 1 0 0 -2 -1 6 -1 2 1 0 0 2 0 0 -1 0 -1 -1 3 -1 0 1 -1 -1 0 1 0 0 -1 -2 0 2 -1 2 0 0 -1 0 1]
 [-3 0 0 1 0 1 -1 1 1 2 -1 -2 1 -2 -1 6 -1 -1 1 1 -2 -2 -1 3 0 -1 -1 -1 1 0 1 1 1 0 2 -1 3 -2 1 3 -1 0 2 1 0 0 -1 0]
 [-2 -3 -3 1 -1 2 1 0 1 1 -1 -1 -1 -1 2 -1 6 0 3 1 1 0 -3 -1 -2 1 -1 -1 0 0 2 -1 0 1 0 1 -1 0 0 -1 0 -2 1 3 1 -1 1 0]
 [-1 -2 -2 0 -3 0 -2 2 -1 3 -2 -2 -3 3 1 -1 0 6 0 2 0 0 -1 -1 1 1 0 0 -1 1 1 1 -2 -2 0 -1 -1 1 1 0 0 -2 1 -1 -1 1 -2 0]
 [-2 -2 -3 1 -2 2 1 1 2 2 -2 0 1 0 1 3 0 6 1 -1 -1 -2 -2 -1 0 0 -1 0 -1 0 -1 0 2 -1 0 0 0 2 1 -1 -2 0 2 -1 0 0 -1]
 [-2 -2 -2 0 -3 3 -2 0 0 2 -3 -1 -1 1 0 1 1 2 1 6 -1 1 -2 -1 -1 0 1 -1 1 -1 0 1 0 0 1 2 1 0 0 1 -1 -2 1 1 -1 1 0 0]
 [0 -2 0 0 1 0 0 -1 -2 -1 1 1 0 0 2 -2 1 0 -1 -1 6 1 1 -1 -2 1 1 0 0 -1 0 0 -2 -1 1 -1 -1 0 -1 -3 1 0 0 0 2 1 1 1]
 [1 0 2 -3 0 1 -2 -1 1 -2 -1 0 0 1 0 -2 0 0 -1 1 1 6 1 0 -1 2 1 0 1 2 -1 2 0 -1 1 0 -1 0 -2 -1 0 1 -1 -1 2 0 1 0]
 [2 3 3 -1 1 -2 -2 -1 1 -3 2 2 1 1 0 -1 -3 -1 -2 -2 1 1 6 1 2 -1 0 0 1 0 -1 1 0 -1 0 -1 0 1 -1 0 2 2 -2 -2 0 0 1 1]
 [-1 1 2 -1 1 -1 -2 0 1 -1 1 1 -1 1 -2 -1 3 -1 -1 -2 -1 -1 0 1 6 2 0 -2 -1 0 1 1 2 1 -1 1 -1 2 0 0 2 1 2 1 0 2 -2 0 0]
 [0 2 0 0 -1 -1 -2 2 1 -1 0 -1 0 1 0 0 -2 1 -1 -1 -2 -1 2 2 6 -1 -2 1 0 0 -1 1 -1 -2 0 1 2 0 2 2 1 1 -1 -1 -1 -1 0]
 [0 -2 1 -2 0 0 0 0 0 0 -1 -1 0 1 -1 -1 1 1 0 0 1 2 -1 0 -1 6 0 -2 0 1 1 2 0 0 0 0 -1 0 0 -2 -2 1 -1 -1 2 0 -1 0]
 [2 -1 0 -2 -1 1 2 0 -1 1 0 0 -1 1 1 -1 -1 -1 0 0 1 1 1 0 -2 -2 0 6 0 -1 0 -2 -1 0 0 -1 1 -1 1 1 0 -2 0 0 -1 0 3 1 -2]
 [2 2 0 1 2 -1 1 0 -2 0 1 0 -1 -1 3 -1 -1 0 -1 -1 0 0 0 -1 1 -2 0 6 -2 0 -1 -1 -1 0 -1 0 0 -1 -2 0 1 1 0 -1 -1 0 1 -1]
 [-1 0 1 -1 0 2 -2 -1 1 -2 -1 1 1 0 -1 1 0 -1 0 1 0 1 1 0 0 0 -1 -2 6 0 0 0 0 0 2 0 1 0 0 -1 0 0 0 -1 1 1 1 1]
 [0 0 2 -2 -1 0 -1 1 1 1 0 -1 -1 -1 0 0 0 1 -1 -1 -1 2 0 1 0 1 0 0 0 6 1 0 1 -2 1 0 -1 1 0 1 -1 0 0 -1 2 1 0 -1]
 [-2 -1 -1 1 0 0 -1 0 0 2 -2 -1 -1 -2 1 1 2 1 0 0 0 -1 -1 1 -1 -2 0 1 -1 6 0 1 0 1 -1 0 -1 -1 -1 -1 1 -1 1 1]
 [-1 0 1 -1 0 -1 -3 1 0 0 -2 -1 1 2 -1 1 -1 1 -1 1 0 2 1 2 1 2 -1 -1 0 0 1 6 -1 -2 0 -2 0 0 -1 0 0 1 0 0 0 -1 -1 1]
 [-1 1 1 1 1 0 1 1 0 3 0 1 0 2 -2 -1 1 0 -2 0 0 -2 0 0 1 -1 0 0 -1 0 1 1 -1 6 0 1 0 1 -1 0 1 -1 0 -1 1 1 -1 0 0]
 [0 0 -2 0 0 2 2 0 1 -1 -1 -1 0 -1 0 0 1 -2 2 0 -1 -1 -1 -1 1 0 0 0 0 -2 0 -2 0 6 -1 3 1 0 -1 1 -1 0 1 1 0 -1 0 -1]
 [-2 -1 1 1 1 1 -1 -1 0 0 0 0 1 -2 1 2 0 0 -1 1 1 1 0 1 -2 0 -1 -1 2 1 1 0 1 -1 6 -1 1 -2 0 0 0 0 1 0 1 0 -1 2]
 [1 0 -1 -1 -1 2 1 -1 0 -1 0 0 -1 -1 0 -1 1 -1 0 2 -1 0 -1 -1 0 0 1 0 0 0 -1 -2 0 3 -1 6 0 1 -1 1 -1 0 0 1 0 0 1 -2]
 [-2 0 0 1 0 1 -2 -1 0 1 -2 -1 1 -3 0 3 -1 -1 0 1 -1 -1 0 2 1 -1 -1 0 1 -1 2 0 1 1 1 0 6 -1 0 2 0 -1 1 0 0 0 0 1]
 [1 -1 0 -2 -2 -1 -1 1 -1 0 0 0 -2 2 -1 -2 0 1 0 0 0 0 1 0 2 0 1 -1 0 1 -1 0 -1 0 -2 1 -1 6 1 0 0 1 -1 -1 0 0 1 0]
 [-1 -2 -1 1 -2 0 1 1 0 2 0 -1 1 1 -2 1 0 1 2 0 -1 -2 -1 0 0 0 1 -2 0 0 -1 -1 0 -1 0 -1 0 1 6 1 -1 -1 0 0 -1 2 -1 0]
 [-2 1 -1 1 -2 1 -1 2 2 2 -1 -2 0 -1 0 3 -1 0 1 1 -3 -1 0 2 2 -2 0 0 -1 1 0 0 1 1 0 1 2 0 1 6 0 -1 2 1 -1 0 -1 -1]
 [0 2 1 1 1 -2 -2 -1 1 -2 2 2 1 0 2 -1 0 0 -1 -1 1 0 2 1 2 -2 1 0 -1 -1 0 -1 0 0 -1 0 6 -1 0 0 -1 -2 0 2]
 [3 2 3 -2 3 -2 1 -1 -1 -3 3 1 0 0 -1 0 -2 -2 -2 -2 0 1 2 2 1 1 0 1 0 0 -2 1 0 0 0 0 -1 1 -1 -1 -1 6 -2 -2 1 -1 1 -1]
 [-3 -1 -3 2 -1 2 0 3 0 2 -2 -2 -1 -1 2 2 1 1 0 1 0 -1 -2 1 1 -1 0 0 0 0 1 0 -1 1 1 0 1 -1 0 2 0 -2 6 1 0 0 -2 0]
 [-3 -2 -3 2 -2 2 0 1 2 1 -1 -2 1 -1 0 1 3 -1 2 1 0 -1 -2 0 -1 -1 -1 -1 -1 1 0 1 1 0 1 0 -1 0 1 0 -2 1 6 0 -1 0 1]
 [-1 -2 1 -2 0 2 0 0 1 -1 0 -1 0 -2 0 0 1 -1 -1 -1 2 2 0 2 -1 2 0 -1 1 2 1 0 1 0 1 0 0 0 -1 -1 -1 1 0 0 6 0 1 0]
 [0 -2 0 -1 -2 2 0 0 -1 2 -1 -1 0 0 -1 0 -1 1 0 1 1 0 0 -2 -1 0 3 0 1 1 -1 -1 -1 -1 0 0 0 2 0 -2 -1 0 -1 0 6 1 -1]
 [2 0 1 -2 1 1 0 -3 0 -2 1 1 0 -1 0 -1 1 -2 0 0 1 1 1 0 -1 -1 1 1 1 0 -1 -1 0 0 -1 1 0 1 -1 -1 0 1 -2 0 1 1 6 -1]
 [-2 -1 0 2 0 -1 -2 0 0 0 -1 1 1 0 1 0 0 0 -1 0 1 0 1 0 0 0 -2 -1 1 -1 1 1 0 -1 2 -2 1 0 0 -1 2 -1 0 1 0 -1 -1 6]]
```

T (8×48):

```
[[5 -6 2 1 0 1 -4 3 1 0 1 1 -1 0 0 -1 -1 -2 0 0 0 -1 -1 -1 -1 -1 -1 0 0 0 1 1 1 1 0 0 0 -2 1 1 1 1 1 -1 -1 -1 0 -1]
 [-1 -2 -3 -2 1 0 0 2 0 -1 2 2 0 -1 0 1 -1 0 0 0 -1 1 1 0 0 1 0 1 0 0 1 0 0 0 0 0 0 0 0 0 0 -1 -1 0 -1 0 0 0]
 [-5 4 1 0 -2 -4 7 -3 0 -3 -3 -3 0 -1 1 1 1 2 -1 2 1 1 1 1 0 0 0 0 1 0 -1 -1 -1 -1 -1 -1 0 2 -1 0 -1 -1 -1 1 0 1 0 0]
 [-5 3 0 0 0 0 2 -4 0 2 -1 -1 0 3 0 -1 0 -1 -1 0 0 0 0 1 1 0 1 0 0 1 0 -1 -1 0 0 0 0 0 0 -1 0 1 0 0 0 0 0 0]
 [11 -4 -2 0 1 3 -8 7 -1 0 4 3 1 -4 -1 1 0 1 3 -2 -1 0 -1 -2 -1 0 -1 -1 -1 -2 0 2 2 0 1 1 0 -1 0 1 0 -1 1 -1 1 0 0 1]
 [1 -2 6 2 1 2 -2 0 1 0 -3 -1 -1 3 0 -2 1 -2 -1 0 1 -3 -1 -1 0 -1 0 0 -1 0 1 -1 0 0 0 0 0 -1 1 1 1 2 1 0 0 -1 0 -1]
 [-7 4 -4 -3 -3 -3 6 -5 -1 0 1 0 0 0 2 3 -1 1 -1 -1 -1 3 1 1 1 1 0 1 1 0 -1 1 0 0 0 0 0 2 0 -2 -1 -2 -1 1 0 1 0 0]
 [-1 -2 3 3 3 -2 2 1 3 0 -1 -3 -1 0 -2 -5 -1 0 0 5 2 -2 1 2 -2 -1 0 0 2 3 -1 -1 -2 1 -2 -1 1 -1 -1 1 0 1 0 0 -1 -1 0 0]]
```

A.2 Dimension 39

Certificate `records/dim39-delta18/P48p-adc3124f`; the ambient is P_{48p} and $\mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}} = \mathbf{K}_9$.

T (9×48):

```
[[5 -6 2 1 0 1 -4 3 1 0 1 1 -1 0 0 -1 -1 -2 0 0 0 -1 -1 -1 -1 -1 -1 0 0 0 1 1 1 1 0 0 0 -2 1 1 1 1 1 -1 -1 -1 0 -1]
 [-5 4 1 0 -2 -4 7 -3 0 -3 -3 -3 0 -1 1 1 1 2 -1 2 1 1 1 1 0 0 0 0 1 0 -1 -1 -1 -1 -1 -1 0 2 -1 0 -1 -1 -1 1 0 1 0 0]
 [-10 7 1 0 -2 -4 9 -7 0 -1 -4 -4 0 2 1 0 1 1 -2 2 1 1 1 2 1 0 1 0 1 1 -1 -2 -2 -1 -1 -1 0 2 -1 -1 -1 0 -1 1 0 1 0 0]
 [0 1 -4 -2 0 -1 1 2 -1 -2 2 1 1 -3 0 2 0 2 1 0 -1 2 1 0 0 1 0 0 0 -1 0 0 0 -1 0 0 0 1 -1 0 -1 -2 -1 0 0 1 0 1]
 [-11 5 -2 -2 -1 -4 9 -5 0 -2 -2 -2 0 1 1 1 0 1 -2 2 2 0 2 2 2 1 1 1 1 1 1 0 -2 -2 -1 -1 -1 0 2 -1 -1 -1 -1 -2 1 -1 1 0 0]
 [1 -1 2 0 1 1 -1 2 0 -2 -1 0 0 0 0 0 1 0 0 0 0 -1 0 -1 0 0 0 0 -1 -1 1 -1 0 -1 0 0 0 0 0 1 0 0 0 0 0 0 0 0]
 [-6 3 -2 -3 -2 -2 5 -3 -1 -2 0 0 0 0 2 3 0 1 -1 -1 2 1 0 1 1 0 1 0 -1 0 0 0 -1 0 0 0 2 0 -1 -1 -2 -1 1 0 1 0 0]
 [5 -8 6 1 0 0 -3 4 2 -3 1 -1 -2 0 1 -2 -1 -2 0 1 0 -2 -1 -2 -2 -2 -2 0 0 -1 1 1 1 0 -1 0 1 -2 1 2 0 0 2 -1 -1 -1 0 -1]
 [-12 3 7 0 -2 -4 10 -7 1 -3 -4 -6 -2 5 2 -2 0 -1 -3 3 1 -1 1 1 0 -1 0 1 1 1 0 -2 -2 -1 -2 -1 1 1 0 0 -1 0 0 1 -1 0 0 -1]]
```

A.3 Dimension 38

Certificate `records/dim38-delta9/P48p-a17f250f`; the ambient is P_{48p} and $\mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}} = \mathbf{K}_{10}$.

T (10×48):

```
[[5 -6 2 1 0 1 -4 3 1 0 1 1 -1 0 0 -1 -1 -2 0 0 0 -1 -1 -1 -1 -1 -1 0 0 0 1 1 1 1 0 0 0 -2 1 1 1 1 1 -1 -1 -1 0 -1]
 [-5 4 1 0 -2 -4 7 -3 0 -3 -3 -3 0 -1 1 1 1 2 -1 2 1 1 1 1 0 0 0 0 1 0 -1 -1 -1 -1 -1 -1 0 2 -1 0 -1 -1 -1 1 0 1 0 0]
 [-10 7 1 0 -2 -4 9 -7 0 -1 -4 -4 0 2 1 0 1 1 -2 2 1 1 1 2 1 0 1 0 1 1 -1 -2 -2 -1 -1 -1 0 2 -1 -1 -1 0 -1 1 0 1 0 0]
 [0 1 -4 -2 0 -1 1 2 -1 -2 2 1 1 -3 0 2 0 2 1 0 -1 2 1 0 0 1 0 0 0 -1 0 0 0 -1 0 0 0 1 -1 0 -1 -2 -1 0 0 1 0 1]
 [-11 5 -2 -2 -1 -4 9 -5 0 -2 -2 -2 0 1 1 1 0 1 -2 2 0 2 2 2 1 1 1 1 1 1 0 -2 -2 -1 -1 -1 0 2 -1 -1 -1 -1 -2 1 -1 1 0 0]
 [1 -1 2 0 1 1 -1 2 0 -2 -1 0 0 0 0 0 1 0 0 0 0 -1 0 -1 0 0 0 0 -1 -1 1 -1 0 -1 0 0 0 0 0 1 0 0 0 0 0 0 0 0]
 [-6 3 -2 -3 -2 -2 5 -3 -1 -2 0 0 0 0 2 3 0 1 -1 -1 2 1 0 1 1 0 1 0 -1 0 0 0 -1 0 0 0 2 0 -1 -1 -2 -1 1 0 1 0 0]
 [5 -8 6 1 0 0 -3 4 2 -3 1 -1 -2 0 1 -2 -1 -2 0 1 0 -2 -1 -2 -2 -2 -2 0 0 -1 1 1 1 0 -1 0 1 -2 1 2 0 0 2 -1 -1 -1 0 -1]
 [-12 3 7 0 -2 -4 10 -7 1 -3 -4 -6 -2 5 2 -2 0 -1 -3 3 1 -1 1 1 0 -1 0 1 1 1 0 -2 -2 -1 -2 -1 1 1 0 0 -1 0 0 1 -1 0 0 -1]
 [-2 1 -5 -2 0 -2 1 -1 -1 0 1 0 1 -2 0 0 -1 1 0 1 -1 1 2 0 -1 1 0 0 1 1 -1 0 0 0 -1 -1 0 0 0 0 0 0 0 0 0 0 0 0]]
```

A.4 Dimension 43

Certificate `records/dim43-delta226/P48m-1f02996f`. The ambient is P_{48m}; $\mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}} = \mathbf{K}$ is the Gram matrix of Section 4 and $\mathbf{W}\mathbf{G}\mathbf{W}^{\mathsf{T}} = [\,0; I_5\,]$. Its Gram matrix $\mathbf{G}$ is the catalogue entry `P48m` [19, 20] reproduced verbatim; the dimension-45 packing below uses the same ambient and the same $\mathbf{G}$.

G (the Gram matrix of P_{48m}):

```
[[6 -2 1 2 -1 -1 -3 1 1 -1 2 -3 2 0 1 -2 -3 -2 2 -2 -3 -2 -1 1 0 2 1 0 -2 2 -2 -2 3 2 1 -2 1 3 3 2 -3 3 0 -3 -2 0 2 -2]
 [-2 8 -1 -2 -2 1 0 -3 -2 3 2 2 2 -2 3 0 -2 3 -4 0 4 4 -3 1 2 1 2 0 2 0 -1 3 -4 -1 -4 2 0 -4 -2 -3 1 0 0 -1 3 3 -4 3]
 [1 -1 6 -2 0 -3 0 2 -2 2 -1 -3 -2 1 -2 2 -3 -2 2 2 -3 -1 -1 0 -1 2 1 2 -3 0 1 -1 0 -1 -1 0 0 0 -2 2 2 -2 2 -1 0 -1 2 2]
 [2 -2 -2 6 1 2 -3 1 0 -3 0 1 1 1 2 -3 1 -2 4 -2 0 -3 1 2 -2 1 0 -1 2 1 1 -2 0 2 2 -2 -1 1 1 0 -1 3 -1 -1 -1 -2 2 -1]
 [-1 -2 0 1 6 -2 -2 2 -2 -2 1 2 -3 -1 -1 0 2 -3 0 1 -1 -3 2 -2 -3 0 -1 2 1 -2 -1 0 0 -2 3 -1 -3 -2 1 2 1 -2 2 2 -2 -3 -1 -2]
 [-1 1 -3 2 -2 6 2 0 1 0 -1 2 1 2 1 0 1 1 1 -2 3 2 0 2 1 -1 1 -2 2 0 1 1 0 0 -2 0 0 1 1 -2 1 2 -3 -1 0 1 1 1]
 [-3 0 0 -3 -2 2 8 -1 2 3 -1 1 0 1 -3 4 2 2 -2 2 3 3 2 1 1 -2 -2 1 -1 -1 0 0 2 -2 -2 0 2 0 -2 0 0 -2 -1 2 0 0 1 0]
 [1 -3 2 1 2 0 -1 6 -1 0 -1 0 -1 0 -1 -1 0 -2 2 1 -2 -3 0 1 -2 -1 -1 0 -1 -1 0 0 1 0 2 -2 0 1 0 1 3 -2 2 0 -3 -3 2 0]
 [1 -2 -2 0 -2 1 2 -1 6 -2 -1 0 2 1 -1 0 1 2 3 0 1 2 2 1 0 -1 -2 0 0 2 -1 0 3 2 2 -1 1 4 2 1 -3 1 -1 0 -1 2 2 -3]
 [-1 3 2 -3 -2 0 3 0 -2 6 1 0 0 0 -1 1 -1 1 -3 2 1 1 -2 1 1 0 1 1 -2 0 0 0 0 -2 -4 1 2 -2 -2 -1 1 -1 1 -1 1 1 -1 3]
 [2 2 -1 0 1 -1 -1 -1 -1 1 6 1 2 -3 2 -1 -1 0 -1 -1 0 0 0 0 1 0 0 2 -1 1 -3 0 0 -1 -1 0 0 -2 1 1 -3 2 1 -1 -1 0 -3 -1]
 [-3 2 -3 1 2 2 1 0 0 0 1 6 0 -1 1 0 3 1 0 0 4 1 2 0 -1 -2 -2 0 3 -1 0 1 -1 -1 0 0 -1 -3 -1 -1 1 -1 0 2 0 -1 -2 0]
 [2 2 -2 1 -3 1 0 -1 2 0 2 0 6 -1 2 -2 -1 2 1 -2 1 2 0 3 1 -1 -1 -1 0 1 -1 0 0 2 0 0 2 1 0 -1 -2 3 0 -1 0 2 -1 -1]
 [0 -2 1 1 -1 2 1 0 1 0 -3 -1 -1 6 -3 1 0 -2 1 0 0 0 0 1 -1 1 2 0 0 1 2 -2 1 0 0 0 0 2 1 1 1 2 -2 -1 0 2 3 1]
 [1 3 -2 2 -1 1 -3 -1 -1 -1 2 1 2 -3 6 -2 -1 2 0 -2 1 1 -2 1 1 1 0 -1 2 0 0 2 -3 1 -1 0 0 -1 0 -2 -1 1 0 -1 1 0 -2 1]
 [-2 0 2 -3 0 0 4 -1 0 1 -1 0 -2 1 -2 6 0 0 -3 1 0 3 1 -1 0 0 0 1 0 -2 0 0 1 -2 -2 1 0 0 -1 1 1 -2 -1 1 1 0 1 1]
 [-3 -2 -3 1 2 1 2 0 1 -1 -1 3 -1 0 -1 0 6 1 2 1 1 -1 3 -1 -1 -2 -3 -1 1 -1 1 0 1 -1 2 0 0 -1 0 -1 0 -1 0 3 0 -1 -1 -2]
 [-2 3 -2 -2 -3 1 2 -2 2 1 0 1 2 -2 2 0 1 6 0 1 2 4 -1 1 2 -1 -1 -1 1 0 0 3 -1 1 -1 2 2 0 -1 -2 -1 -1 0 1 2 2 -2 1]
 [2 -4 -2 4 0 1 -2 2 3 -3 -1 0 1 1 0 -3 2 0 8 0 -2 -2 1 2 -2 0 -1 -1 1 3 0 -1 2 4 3 -1 1 3 1 0 -1 3 -1 0 -1 -1 2 -2]
 [-2 0 2 -2 1 -2 2 1 0 2 -1 0 -2 0 -2 1 1 1 0 6 1 0 0 1 -1 1 -1 3 0 0 0 0 0 -1 1 -1 0 -1 -2 1 1 -3 2 1 0 0 1 0]
 [-3 4 -3 0 -1 3 3 -2 1 1 0 4 1 0 1 0 1 2 -2 1 8 2 0 1 1 -1 -1 0 3 0 0 1 -2 -1 -1 -1 0 -2 -2 -2 0 -1 -1 1 0 1 -1 0]
 [-2 4 -1 -3 -3 2 3 -3 2 1 0 1 2 0 1 3 -1 4 -2 0 2 8 -1 1 2 0 0 0 2 0 -1 3 -1 0 -3 3 1 0 0 -1 0 0 -2 0 2 4 -1 2]
 [-1 -3 -1 1 2 0 2 0 2 -2 0 2 0 0 -2 1 3 -1 1 0 0 -1 6 0 -1 -2 -3 1 0 -1 1 -2 2 -1 2 -1 -2 0 0 2 -1 0 0 2 -1 -2 1 -3]
 [1 1 0 2 -2 2 1 1 1 1 0 0 3 1 1 -1 -1 1 2 1 1 1 0 6 -1 1 0 1 1 1 0 0 0 1 0 -1 1 1 -1 0 0 2 0 -1 0 0 2 1]
 [0 2 -1 -2 -3 1 1 -2 0 1 1 -1 1 1 0 -1 2 -2 -1 1 2 -1 -1 6 -1 1 -2 -1 1 0 0 -1 0 -2 1 1 -1 0 -1 -1 1 -2 -1 1 2 -1 1]
 [2 1 2 1 0 -1 -2 -1 -1 0 0 -2 -1 1 1 0 -2 -1 0 1 -1 0 -2 1 -1 6 2 2 0 1 0 0 0 0 -1 0 -1 0 1 1 -1 1 0 -2 1 1 1 0]
 [1 2 1 0 -1 1 -2 -1 -2 1 0 -2 -1 2 0 0 -3 -1 -1 -1 0 -3 0 1 2 6 0 0 1 0 0 -1 0 -2 1 -1 0 1 0 1 2 -1 -3 1 2 0 2]
 [0 0 2 -1 2 -2 1 0 0 1 2 0 -1 0 -1 1 -1 -1 -1 3 0 0 1 1 -2 2 0 6 -1 0 -1 0 0 -2 0 -1 -2 -1 0 3 -1 -1 2 0 -1 0 0 -1]
 [-2 2 -3 2 1 2 -1 -1 0 -2 -1 3 0 0 2 0 1 1 1 0 3 2 0 1 -1 0 0 -1 6 -1 0 1 -2 1 1 0 -1 -1 0 -1 1 0 -2 1 1 1 0 0 1]
 [2 0 0 1 -2 0 -1 -1 2 0 1 -1 1 1 0 -2 -1 0 3 0 0 0 -1 1 1 1 1 0 -1 6 -1 -1 0 2 0 0 1 1 0 0 -2 3 -1 -2 0 2 1 0]
 [-2 -1 1 1 -1 1 0 0 -1 0 -3 0 -1 2 0 0 1 0 0 0 0 -1 1 0 0 0 0 -1 0 -1 6 -1 -2 0 -1 1 -1 0 -2 -1 2 0 0 0 2 0 1 2]
 [-2 3 -1 -2 0 1 0 0 0 0 0 0 1 0 -2 2 0 0 3 -1 0 1 3 -2 0 0 0 0 0 1 -1 -1 6 -2 -1 -1 2 0 -1 0 -2 2 -2 1 1 1 1 -3 1]
 [3 -4 0 0 0 0 2 1 3 0 0 -1 0 1 -3 1 1 -1 2 0 -2 -1 2 0 -1 0 -1 0 -2 0 -2 -2 8 0 1 -2 1 4 4 2 -2 1 -1 -1 -2 -1 3 -3]
 [2 -1 -1 2 -2 0 -2 0 2 -2 -1 -1 2 0 1 -2 -1 1 4 -1 -1 0 -1 1 0 0 0 -2 1 2 0 -1 0 6 1 0 2 2 0 0 -1 2 -1 0 0 0 2 0]
 [1 -4 -1 2 3 -2 -2 2 2 -4 -1 0 0 0 -1 -2 2 -1 3 1 -1 -3 2 0 -2 -1 -2 0 1 0 -1 -1 1 1 8 -3 -1 2 1 2 -1 -1 2 2 -2 -1 1 -4]
 [-2 2 0 -2 -1 0 0 -2 -1 1 0 0 0 0 0 1 0 2 -1 -1 -1 3 -1 -1 1 0 1 -1 0 0 1 2 -2 0 -3 6 1 -1 -1 -1 1 0 -1 1 2 2 -2 2]]
```

```
[1 0 0 -1 -3 0 2 0 1 2 0 -1 2 0 0 0 0 2 1 0 0 1 -2 1 1 -1 -1 -2 -1 1 -1 0 1 2 -1 1 6 1 -1 -1 -1 0 0 1 0 0 0 1]
[3 -4 0 1 -2 1 0 1 4 -2 -2 -3 1 2 -1 0 -1 0 3 -1 -2 0 0 1 -1 0 0 -1 -1 1 0 -1 4 2 2 -1 1 8 3 1 -2 1 -1 -2 -1 1 4 -2]
[3 -2 -2 1 1 1 -2 0 2 -2 1 -1 0 1 0 -1 0 -1 1 -2 -2 0 0 -1 0 1 1 0 0 0 -2 0 4 0 1 -1 -1 3 8 2 -3 2 -1 -2 -2 2 1 -2]
[2 -3 2 0 2 -2 0 1 1 -1 1 -1 1 -2 1 -1 -2 0 1 -2 -1 2 0 -1 1 0 3 -1 0 -1 -2 2 0 2 -1 -1 1 2 6 -2 -1 1 0 -3 -1 3 -2]
[-3 1 2 -1 1 1 0 3 -3 1 -3 1 -2 1 -1 1 0 -1 -1 1 0 0 -1 0 -1 -1 1 -1 1 -2 2 2 -2 -1 -1 1 -1 -2 -3 -2 8 -2 0 1 1 -1 0 3]
[3 0 -2 3 -2 2 -2 -2 1 -1 2 -1 3 2 1 -2 -1 -1 3 -3 -1 0 0 2 1 1 2 -1 0 3 0 -2 1 2 -1 0 0 1 2 -1 -2 8 -3 -3 1 2 0 0]
[0 0 2 -1 2 -3 -1 2 -1 1 1 0 0 -2 0 -1 0 0 -1 2 -1 -2 0 0 -2 0 -1 2 -2 -1 0 1 -1 -1 2 -1 0 -1 -1 1 0 -3 6 1 -1 -1 -2 -1]
[-3 -1 -1 -1 2 -1 2 0 0 -1 -1 2 -1 -1 -1 1 3 1 0 1 1 0 2 -1 -1 -2 -3 0 1 -2 0 1 -1 0 2 1 1 -2 -2 0 1 -3 1 6 0 -2 -2 -1]
[-2 3 0 -1 -2 0 0 -3 -1 1 -1 0 0 0 1 1 0 2 -1 0 0 2 -1 0 1 1 1 -1 1 0 2 1 -2 0 -2 2 0 -1 -2 -3 1 1 -1 0 6 2 -2 3]
[0 3 -1 -2 -3 1 0 -3 2 1 0 -1 2 2 0 0 -1 2 -1 0 1 4 -2 0 2 1 2 0 0 2 0 1 -1 0 -1 2 0 1 2 -1 -1 2 -1 -2 2 8 -1 0]
[2 -4 2 2 -1 1 1 2 2 -1 -3 -2 -1 3 -2 1 -1 -2 2 1 -1 -1 1 2 -1 1 0 0 0 1 1 -3 3 2 1 -2 0 4 1 3 0 0 -2 -2 -2 -1 8 0]
[-2 3 2 -1 -2 1 0 0 -3 3 -1 0 -1 1 1 1 -2 1 -2 0 0 2 -3 1 1 0 2 -1 1 0 2 1 -3 0 -4 2 1 -2 -2 -2 3 0 -1 -1 3 0 0 8]]
```

T (5×48):

```
[[-1 1 0 0 0 0 0 0 -1 0 0 -1 0 1 1 0 0 -1 0 1 0 0 1 0 0 -1 0 0 0 0 0 0 1 1 1 0 0 0 0 0 0 -1 0 0 0 0 0 0 0 0]
 [-1 -2 7 -2 5 2 -2 -6 1 4 1 2 3 -1 -1 0 1 0 4 -2 1 0 0 1 2 1 1 -1 -1 -2 0 1 -2 1 1 -2 0 1 1 -2 0 -2 -2 0 -1 0 1 0]
 [-1 0 -7 1 -3 -3 -1 6 0 -1 3 -5 -5 1 0 0 0 1 -2 -2 2 4 1 1 -3 0 0 0 -1 1 1 -2 1 0 0 0 1 -2 -1 1 1 0 1 -1 2 0 0 -1]
 [2 -2 -2 1 1 0 1 -2 -3 0 -4 3 2 0 -1 2 -3 3 0 1 -1 -3 1 -1 1 1 -2 0 0 1 -2 1 -1 -1 -1 0 0 1 0 1 1 1 1 -1 -1 1 -1 1]
 [-1 3 -8 1 -8 -2 -3 7 -2 -3 1 -5 -6 0 -1 1 2 -1 -3 -2 2 4 0 1 -3 -1 -1 3 -1 2 2 -2 3 0 1 1 1 -2 -1 2 0 0 1 -1 1 -1 -1 -1]]
```

W (6×48):

```
[[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
 [9 7 1 0 8 6 0 -5 2 4 -4 -2 2 -2 -2 3 1 0 5 0 -1 -1 -3 -1 2 -2 -4 2 0 0 2 -1 -1 -1 0 1 -2 1 0 1 1 -1 1 1 0 -1 0 1]
 [6 8 8 -1 12 8 -1 -8 2 6 -6 0 5 -2 1 3 2 -1 7 0 -1 -3 -2 -2 4 -2 -3 2 0 -1 0 0 -1 0 0 0 -1 3 0 1 0 0 0 1 -1 0 0 1]
 [8 6 2 0 6 5 0 -3 4 3 -2 -2 1 -1 0 2 2 -1 4 1 0 0 -2 -1 1 -2 -2 1 0 -1 1 -2 -1 -1 0 2 -2 0 0 1 1 -1 1 0 1 -1 0 1]
 [8 9 -1 2 6 3 1 -2 2 4 -4 -4 1 0 -1 4 2 0 5 -1 -1 -1 -2 -2 2 -4 2 0 0 1 -1 -1 -1 0 1 -2 1 0 1 1 -1 1 1 0 -1 0 1]
 [8 4 -2 1 4 4 -2 -2 1 2 -3 -2 2 -2 -4 4 2 1 2 0 0 0 -3 0 1 -1 -3 2 0 1 2 0 -1 -1 0 1 -2 0 0 1 1 -1 1 1 0 -1 0 1]]
```

A.5 Dimension 45

Certificate `records/dim45-kissing7379838/P48m-d914e7d8`. The ambient is again P_{48m}, so $\mathbf{G}$ is the matrix printed above. As in dimension 43, the basis encoded by $\mathbf{T}$ is the one whose Gram matrix is the $\mathbf{K}$ of Section 5.3, so $\mathbf{W}\mathbf{G}\mathbf{T}^{\mathsf{T}} = [\,0; I_3\,]$: the cluster is $\{0\}$ together with the dual basis m_1, m_2, m_3, and no separate cluster matrix is needed.

T (3×48):

```
[[4 0 -6 0 1 -1 1 0 0 -2 0 0 -1 1 -1 1 -2 2 -1 1 -1 0 -1 1 -1 0 -1 -1 -1 1 1 -1 0 -1 -1 0 -1 -1 -1 0 1 -1 1 0 0 0 0 1]
 [2 0 6 -1 2 2 3 -3 1 0 -2 4 3 -1 1 -2 1 -2 1 2 -2 -2 -1 -2 2 0 1 0 2 -1 -1 1 -1 0 0 1 -1 2 0 -1 -1 1 0 1 -1 0 0 1]
 [-3 0 -3 2 -3 0 -1 2 0 -1 1 -1 -1 0 -1 1 -2 0 -2 1 -1 0 0 0 -1 0 -1 0 0 1 1 0 1 0 0 0 1 -1 0 1 0 0 0 -1 0 0 -1 -1]]
```

W (4×48):

```
[[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
 [1 1 -7 8 2 -7 0 5 1 4 -4 -4 1 2 1 5 -1 1 0 -1 3 3 0 -2 2 -1 0 1 -2 2 1 1 1 -1 0 1 -1 -1 0 2 1 0 1 0 1 -1 -1 0]
 [0 3 -1 6 5 -3 2 4 2 4 -4 -4 1 3 3 3 0 -1 -1 -1 1 0 0 -3 4 -2 -1 3 2 2 0 2 3 1 0 2 -1 0 -1 1 -1 1 1 0 0 0 -2 0]
 [1 5 -10 12 -1 -10 4 9 0 3 -6 -6 -3 3 2 5 -3 -1 -2 -1 2 1 1 -3 2 -3 -2 3 1 3 1 2 4 0 0 3 -1 -1 -1 3 -1 2 2 -1 1 0 -3 0]]
```

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